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SLO3 Communicate Effectively

Team Members

Most of the time we were able to communicate pretty well with each other, because we all think-alike. Most of the language we use is the same for each of us so it isn't hard to contribute ideas to the team. Although, this doesn't mean everything was smooth sailing.

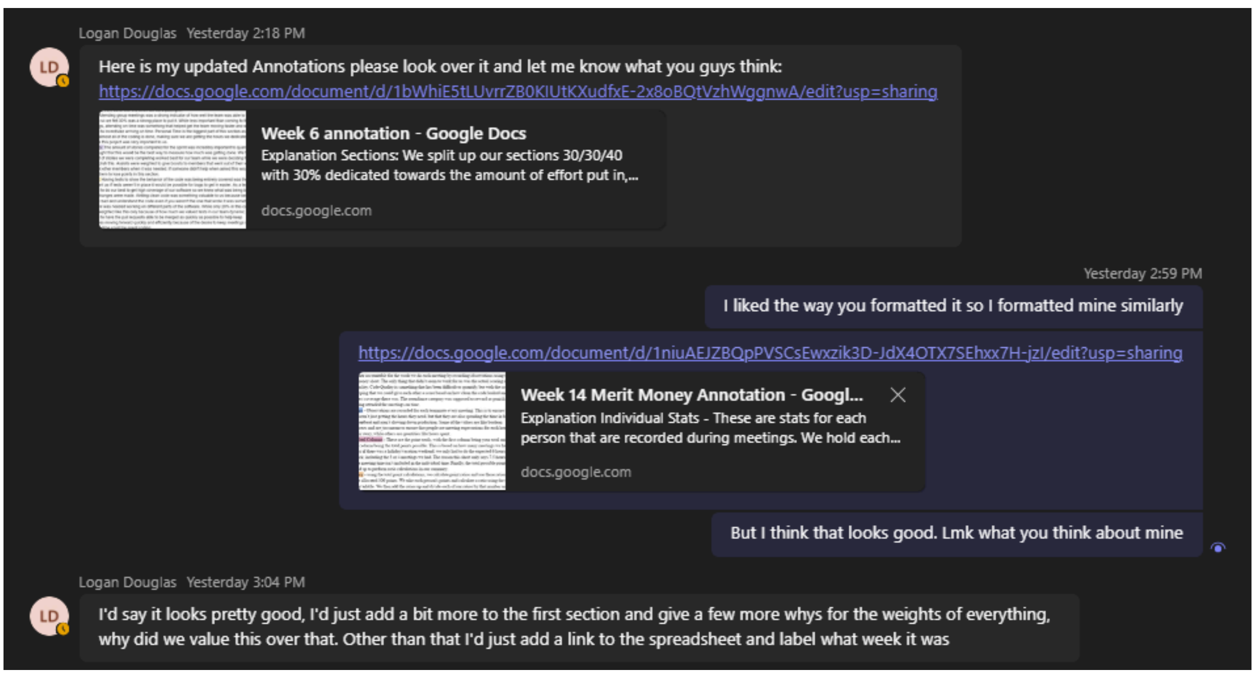
There were times that we failed to communicate our goals/ideas for the app and so it seemed like we had differences of opinion when we were both thinking the same thing or one of us really did have a different idea, but it wasn't clear to the other team members what their idea was.

One trick for communicating was getting more familiar with the domain of our app so that common terms used for that domain wouldn't throw one of the team members off. For this app, the domain was stocks and stock monitoring. When we had first started the project, there was confusion on what the differences were between a stock watchlist, a collection, and a screener profile.

We eventually realized that there was no need for the collection or watchlist after we had carefully looked at our domain. Users were given the ability to make a sort of collection/watchlist-like profile by applying symbol filters, which simplified the app quite a bit.

We had regular team meetings every day to not only ensure that the work was getting done, but that we were all on the same page and had an agreed upon design for our app. Leaders were able to assign stories and would make sure that everyone understood the criteria for their story. Each team member was given the time during our meeting to ask questions if they were uncertain on any of the criteria.

We also had a Microsoft Teams chat that was open for team members to ask questions if they had any more questions after having worked on their story. Oftentimes, we would find issues after we started working on the story, so it was important that we had constant communication through messaging and team heartbeats.



Product Owner

Most of our communication struggles were with our product owner, our course instructor. Because he was our instructor, he was a lot more lenient, but we still had to choose our words carefully around the product owner, because certain terms could leave the product owner concerned.

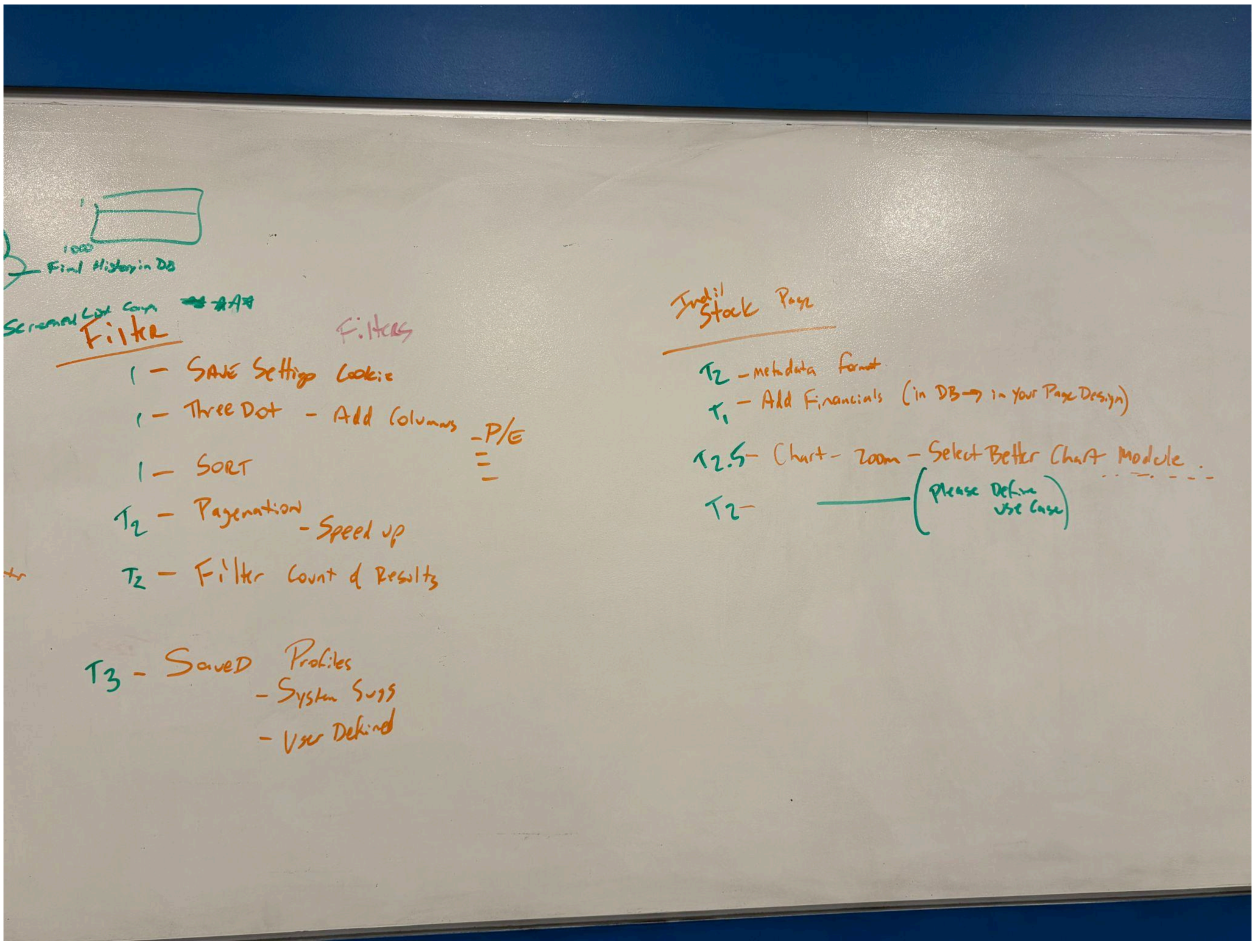
Refactoring and maintenance weren't as important to the product owner as new features were, which also made us careful with our sprint planning that not only would time be spent refactoring, but we would still be adding new helpful features to the app. We would often have to assure the product owner that production on the app wasn't stopped.

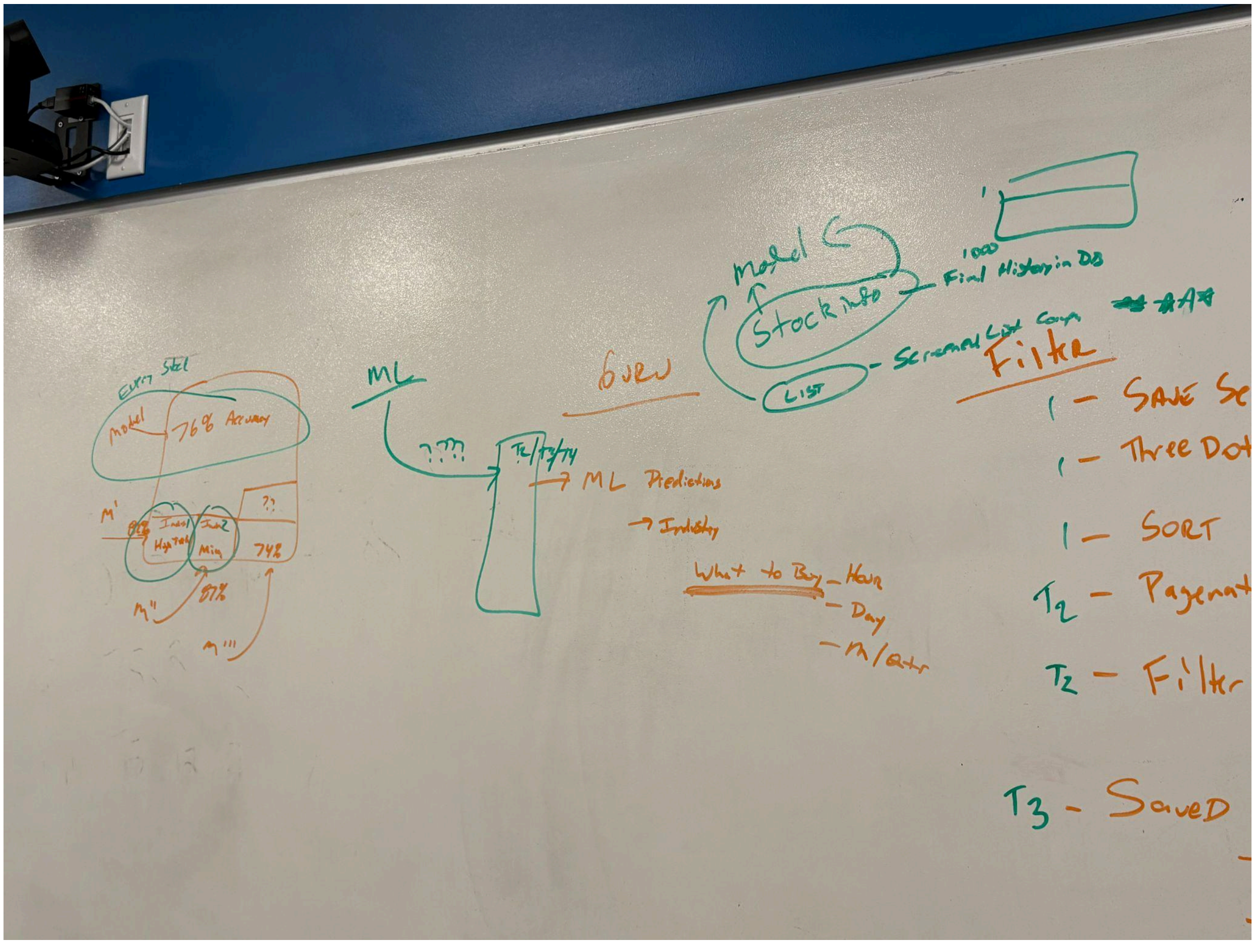
Another important issue was making sure that we actually understood the product owners requirements. We were partly product owners so that also included understanding what each of us wanted for our app. There were times when we did get the wrong idea from the product owners, and sometimes the features we added weren't as good as we thought they would be.

To help clear this up, we would ask each other questions as well as the instructor and take notes of the new requirements. What was nice about this approach, is we could look at this list later and make requirements for our app for the next epics and sprints. This way everyone was on the same page and we all had a clear direction for where we wanted to go next with our app.

We would also give demos of our app so our instructor would know what features we added and how they work. Sometimes we would receive positive feedback and he would leave us with more features to add on top of those features. Other times, we would receive feedback for improving a feature that we would add into our next sprint.

This constant feedback loop helped us shape our app into something more useful and had also helped us also improve relations with the product owner. Of course, we couldn't just rely on instructor feedback. We needed another source of feedback.





Users

This semester, each team member had the opportunity to conduct user tests at least once. Other people were able to find faults in our app that we weren't able to see at the moment. We could see from the user's perspective points of confusion. For example, most people that we interviewed felt like they needed a help page, and so we added a help page with information on how to use the screeners and different parts of our app.

One challenge with user testing is that as the developer you have a bias and sometimes you can forget that the tests are to hear user feedback. For example, sometimes we would ask them very specific questions like “Do you think this button should be here?” and the users would just agree to agree. We made sure to ask users for suggestions on how to make the UI better. It was when we received good, genuine feedback that we knew that we were on the right track.

One tactic that was helpful was to ask the users to take the wheel and complete a certain objective in your app. For one interview, we asked “if you are new to stocks and didn’t know what any of these columns were, where would you go to find information on them?” You would then give the users time to explain their thought process and walk you through how they think they would do it. It was also important to tell the users that there are no dumb questions and that they are not incompetent for not knowing how to use the app. All this would tell us is that there is something wrong with the design of our app that affects the user experience.

We would conduct user testing after Epics when we have new features to demo. We would usually try to find only one person to do it with, so it is only a one-on-one conversation. The team member conducting the interviews would introduce themselves and the app and let them know the purpose of the app. They would also assure them that there are no stupid questions or answers and that the purpose of this was to reveal flaws in our app’s design. We didn’t want to make anyone feel uncomfortable. The interviewer would have them walk them through the use of different features of the app. Finally, at the end of every interview we would ask them for suggestions and features that they would like to see.

We had multiple different trials of this and were able to improve the look and feel of our app over multiple rounds of testing.

ActionableItems.md

UserTesting > ActionableItems.md > ## Items

```
## Items

control z for back
draw a design for inverted screener
try to figure out the bugs
```

Group1.md

UserTesting > Group1.md > # Questions

```
# Questions
Name(s): Neils
1. How familiar are you with stock investing/trading

not farmiliar
2. How interested are you in stock investing/trading

quite interested
3. How does the overall color, font, and layout scheme look? is there anything too jarring or unnatural looking?

overall looks professional maybe a logo and starker contrasts overall color looks good.
4. What tools, explanations, or else would you like in an investment advisor at your familiarity level?

risk level with each investment.
% change
page for benefits of app.
5. What do you like?

liked the structure
hierarchy and navigation
liked switching between stock and candlestick.
6. What could we do better?

search suggestions
7. What bugs did you see?

one that liked logged him out
8. what elements functioned different to how you first expected them to?

didn't understand collections
```

Specialty Group

We looked to our machine learning professor, Michael Olson, as our specialty group for guidance with training and testing the model. It was important that we understood how the neural network was supposed to work, but that first required a business understanding of the objective. At times we didn't think we fully understood what the goal of this was, but we eventually learned that we wanted this feature to be educational and not for only predicting the future.

Our machine learning professor was able to understand better once we had told him our new business understanding of this requirement. Because of this, he was able to give us advice on what to do with our model. He recommended that we don't get too fixated on the accuracy and aim for fifty percent accuracy, because this would still be pretty good, especially for an educational feature.

While this helped, we still needed to be careful with how we communicated with Professor Olson, because he might get the wrong idea about our modeling process. One time, Chris and Nate were explaining our process, and he saw a step that looked incorrect. He then proceeded to explain what was wrong with our process.

The 1-dimensional CNN's take 2-dimensional data and flattens it into 1-dimensional data. Our professor had thought that we had tried to fit 1-dimensional data to the neural network instead of using 2-dimensional data. We took a moment to listen and to think about what he was saying and were then able to clarify that the data we were fitting the model with was not 1d, but we had taken it from a database table with multiple columns and rows.

Once we told him that, we were all on the same page. He was then able to offer us better advice for improving the model. To better communicate with Professor Olson, we needed to gain a business understanding of the problem, perform research and experiments with our data, and come up with the right questions to ask before we met with him.

Failure to do this would end up wasting both of our time, because the Professor wouldn't know where we were stuck and how to get us unstuck, and we wouldn't get as much from our meeting with him. We made sure to come prepared when meeting with him so the time was used effectively.

0. Clearly state Objectively

1. Run 5x24 + ID

2. Run 5x24 - ID

3 Try simpler DNN